Geospatial Time-Machine: Returning to the Past with a Multispectral Temporal Generative Model

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***Abstract*—Data imputation techniques have significantly advanced object detection, prediction, and classification tasks. Despite the vast wealth of available data nowadays, several datasets remain incomplete or expensive to acquire. For instance, collecting temporal UAV imagery of agricultural crops and tracking animal trajectories demands substantial effort and timely execution. Recognizing the pivotal role of temporal data in pattern recognition and scene reconstruction, we introduce an innovative approach to augment temporal datasets. Our proposed methodology utilizes a combination of Graph Neural Network (GNN) and Generative Adversarial Network (GAN) architectures to generate comprehensive synthetic temporal data encompassing multivariate time series. The results demonstrate that imagery generated through backcasting can enhance the accuracy of downstream classification tasks by up to 53% in plant disease detection, particularly in the initial stages of analyzing a crop-growing multi-spectral and multi-temporal dataset.**

***Index Terms*—Generative artificial intelligence, generative adversarial networks, graph neural networks, multispectral, deep learning, data augmentation.**

I. Introduction

In recent years, data imputation and augmentation techniques have witnessed substantial advancements, revolutionizing object detection, prediction, and classification domains. Despite the ever-expanding volume of available data, a persistent challenge continues in incomplete or challenging-to-obtain datasets. A striking example of this challenge can be found in remote sensing, where collecting temporal Unmanned Aerial Vehicle (UAV) imagery of plant growth or tracking the intricate trajectories of wildlife demands a significant investment of effort, time, and resources.

Moreover, while considerable effort has been devoted to providing and enhancing data augmentation techniques [1], initiatives addressing the temporal dimension and complex data, such as multispectral imagery, for generating synthetic data are still in their early stages. A growing body of evidence highlights the significance of incorporating more comprehensive temporal data to enhance performance in time series

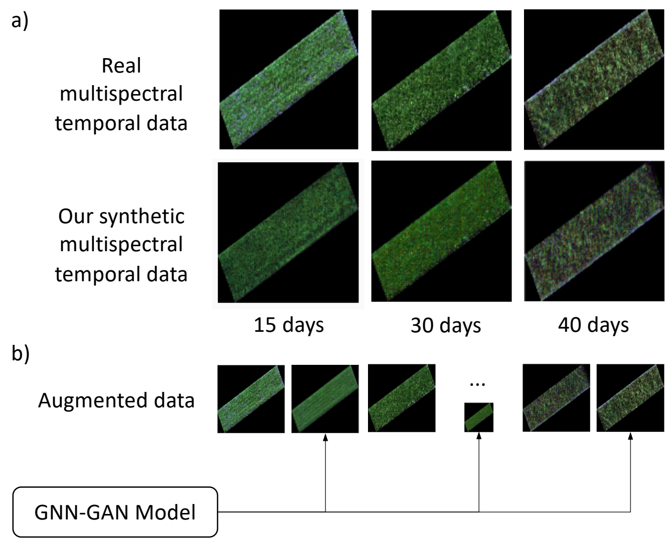

Fig. 1. In a), there is a comparison between real imagery of temporal plant growth data and synthetic samples generated from our GNN-GAN model. In b), the illustration shows the possible outputs from generating samples for other instances of datasets with limited temporal data.

prediction, object detection, or classification tasks [2]. Similar importance is observed for multispectral data [3]. Additionally, the majority of the literature tends to focus more on predictive forecasting (e.g., land coverage [4]), and data fusion [5]–[7]. However, historical temporal data plays a pivotal role in enhancing the accuracy of prediction models [8].

This importance was also acknowledged by Shen *et al.* (2015) [9] presenting a comprehensive review of scene reconstruction techniques, highlighting the significance of temporal-based techniques when reconstructing missing information and performing prediction or classification tasks. Given the current advancements in Generative Artificial Intelligence (GenAI) and the high cost associated with real-time monitoring using remote sensing techniques (e.g., multiple UAV flights for monitoring crops), coupled with the opportunities for innovation in scene reconstruction, as well as the critical role of temporal data in enhancing pattern recognition and analytics, this work raises the question: *could a GenAI-based method generate synthetic multispectral temporal data, imitating real imagery, reconstructing the past of a study area?*

To answer this question, we introduce an innovative approach to augment multispectral and temporal datasets. This approach involves synthetic data generation based on multivariate time series, opening up new avenues for improved analysis and prediction models utilizing synthetically recon-

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structed data.

In this way, we rely on a hybrid model that combines a Graph Neural Network (GNN) with a Generative Adversarial Network (GAN) to accomplish the following: i) capture the profile of feature variation (e.g., vegetation indices) according to temporal snapshots; ii) utilize such variation profile as a discriminator for a multispectral GAN model; and iii) generate novel synthetic multispectral imagery corresponding to specified past temporal instances.

We first utilize image segmentation to convert multispectral images into graphs to obtain the profile of feature variation and leverage the benefits of a graph-based approach, representing spatial and spectral information. Then, we perform a stacking technique for temporal datasets, resulting in multiple temporal graphs. Such methodology resulted in a temporal multispectral GNN-GAN for generating images following back and forecasting results (cf. Figure 1). Through the synergy of a GNN and GAN, we offer a novel approach to address the challenges of temporal data scarcity, ultimately facilitating more robust and accurate analysis in data-driven applications. This approach can be likened to a Geospatial Time-Machine (GTM), offering novel opportunities for comprehensive exploration and understanding of spatiotemporal phenomena. The summarized main contributions of our work are as follows:

1) Seamless integration of a GAN and GNN architecture for multispectral and temporal data augmentation, leading to an integration that significantly bolsters the accuracy and coherence prediction models trained with a combination of real and synthetic data.
2) Introduction of a dual-generator strategy coupled with a novel discriminator designed to evaluate predicted multispectral temporal graphs generated by a pre-trained GNN model. Such discriminator critically assesses the disparities between predicted graphs at an instant $t \pm 1$ and the corresponding real graphs, enhancing the model's grasp of temporal relationships in synthetic data generation.
3) Experimental validation of our enhanced GNN-GAN framework showcasing superior performance in generating precise multispectral temporal graphs. The model maintains temporal coherence and accuracy metrics, surpassing previous approaches and affirming its efficacy in producing accurate synthetic multispectral temporal data.

The rest of this article is organized as follows. Section II compares the proposed GNN-GAN integration with related work. In Section III, the proposed model is described in detail. Section IV provides the description of the dataset and experimental results. Finally, sections V and VI discuss the findings and conclude this article, respectively.

II. Related Work

Several studies have explored the fusion, data imputation, scene reconstruction, and utilization of multispectral data in remote sensing applications, demonstrating advancements in various methodologies. Notably, these investigations encompass diverse approaches ranging from spectral-temporal response surfaces (STRSs) to generative models for image fusion and predictive modeling. However, the incorporation of time-series-based imputation of synthetic imagery for enhancing prediction accuracy is not universally addressed across these studies.

A. Multispectral Temporal Imputation and Prediction Enhancement

The study by Gevaert et al. [10] focuses on constructing STRS using Bayesian theory to fill in missing spectral data within multispectral imagery. This approach significantly enhances correlation (r = 0.953) and decreases RMSE (0.032) regarding field spectral reflectance measurements. It emphasizes the impact of imputation techniques on vegetation indices like OSAVI and MCARI, underscoring the importance of temporal imputation in precision agriculture.

In contrast, Xu et al. [11] concentrate on travel trajectory prediction without explicitly integrating temporal imputation techniques. Their study emphasizes the fusion of multispectral observations and functional land use for travel prediction, highlighting geospatial artificial intelligence (GeoAI) methods while omitting discussions on temporal imputation strategies. Similarly, Azarang et al. [6] explore a generative model for multispectral image fusion but do not specifically address temporal time-series-based imputation strategies. Their focus remains on unsupervised learning and multi-objective loss functions for image fusion, excluding considerations of temporal data.

Using GAN-based approaches, Jiang *et al*. [7] introduce a heterogeneously integrated framework utilizing deep residual cycle generative adversarial networks for image fusion across spatial, temporal, and spectral domains. While tackling fusion challenges comprehensively, their work does not explicitly tackle temporal imputation to enhance predictive capabilities. Yang et al. [12] present the YPM-GAN-GAT model, leveraging GNN and GAN models for corn yield prediction. While resembling the current proposal, they diverge by emphasizing trait attribute completion for yield estimation, not considering temporal imputation and imagery generation approaches – elements pivotal in the current paper's scope. In a similar direction, TimeGAN [13] and GT-GAN [14] offer capabilities for synthesizing data based on predefined window sizes. These methodologies cater to time-series analysis, particularly emphasizing extensive window sampling spanning days, weeks, or even months. They also highlight the necessity for additional temporal data, while not extending their scope to encompass generating synthetic imagery or addressing scenarios with low temporal resolution.

B. Multi-Temporal Data and Scene Reconstruction in Remote Sensing

Prior research has underscored the pivotal role of harnessing comprehensive temporal sequences within multispectral data, significantly enhancing classification and regression tasks and ultimately leading to more precise predictions.

III. Method

A. Problem Statement

In this section, we formalize the problem of generating temporal multispectral synthetic imagery.

Consider a set of multispectral imagery captured at different times, denoted by $\{I_1, I_2, ..., I_T\}$, where T represents the total number of times. The objective is to represent these multispectral images as graphs, transforming the temporal information into multiple graph structures. These graphs will serve as input to a temporal predictive GNN, which aims to learn the temporal relationships between an instant at time t and the subsequent instant at time $t+1$.

As illustrated in Figure 2, X and X' represent the real input multispectral temporal images and the potential synthetic multispectral temporal images, respectively. Notably, multispectral temporal images, particularly in remote sensing and vegetation analysis, exhibit covariance between the last two spectral bands, such as Near Infrared (NIR) and Red-Edge (RE). This covariance is a crucial aspect of these images. Therefore, if covariance exists between the last two bands of X, X' must reproduce the same covariate behavior in its NIR and RE pixel values, which form the inputs for our covariance calculation. This approach compels the latent space's randomness to adhere to the established physical constraint governing the relationship between NIR and RE [15].

In this context, for each input/source image x belonging to X, its covariance coefficients c should also manifest in any corresponding x' element belonging to X'. To achieve this, a generative multispectral model aims to train a generator denoted as $G(z; \theta_h)$, with z representing the input noise vector and θ_h signifying the generator's parameters. The objective of G lies in generating the elements of X' while adhering to the coefficients of C and undergoing evaluation by discriminators D. For each generated X', three discriminators, namely $D_1(X; X'; \theta_1)$, $D_2(X_{\text{spectral}}; X'_{\text{spectral}}; \theta_2)$, and $D_3(G(X); G(X'); \phi)$, are responsible for assessing the distribution, covariance, and graph distance of X' concerning X, respectively. Here, X_{spectral} and X'_{spectral} refer to the spectral profile of X and X', and θ_1, θ_2, and ϕ denote their respective parameters.

B. Multispectral Temporal GNN-GAN Model

The model proposed in this paper consists of seven main modules: optimizer, spectral regularization, two generators, and three discriminators, as depicted in Figure 2. To facilitate the comprehension of our model as well for notation simplicity, consider X and X' vectors $(N, N, 5)$, in the sense that 5-band multispectral imagery will be considered, and N is the imagery size.

Generators. Our model consists of two generators: $G_{\text{multispectral}}$ and G_{temporal} (cf. Generators A and B in Figure 2). $G_{\text{multispectral}}$ takes random noise n as input, with latent space values adjusted by the optimizer O, and generates synthetic multispectral imagery $x'_{\text{multispectral}}$. It aims to produce samples that deceive the discriminators into classifying $x'_{\text{multispectral}}$ as real. Following the architecture of DCGAN [16], $G_{\text{multispectral}}$ employs convolutional layers to transform n into synthetic imagery and utilizes deconvolutional layers to upsample the vector n into a higher-dimensional representation.

G_{temporal} specializes in handling temporal data, following a design similar to $G_{\text{multispectral}}$. It takes noise input to generate synthetic sequences x'_{temporal} specifically suited for temporal aspects. The architecture employs customized convolutional and deconvolutional layers tailored for processing temporal data, aiming to create fid temporal sequences. It's important to note that the output from G_{temporal} is further transformed into a graph format and evaluated by the pre-trained GNN model.

Discriminators. For each set of synthetic samples X', our GNN-GAN model incorporates three discriminators, each serving distinct roles: i) assessing the proximity of each $x' \in X'$ to the real data x, ii) evaluating the spectral divergence between x and x', and iii) considering the temporal graph predictions generated by a pre-trained GNN model for x' at time $t \pm 1$. The discriminator $D_1(x; x'; \theta_s)$ aims to determine the likelihood that x' originates from the real data distribution rather than being synthetically generated. Simultaneously, $D_2(x; x'; \theta)$ receives the spectral information of x and x', assessing the dynamic spectral distance to identify if x_i belongs to the spectral profile of x. Meanwhile, the discriminator $D_3(x'; G(x); \phi)$ critically evaluates the temporal graph prediction $(G(x))$ of the synthetic data x' generated by the GAN's generator, emphasizing the discrepancy between the predicted multispectral temporal graph at time $t \pm 1$ and the real graph at the same time frame.

Thus, the objective of our model with two generators and three discriminators is to find equilibrium by solving the following minimax problem:

$$\begin{aligned}\min_h \max_s & V(D_1, D_2, D_3, G_1, G_2) = \\ & \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D_1(x) + \log D_2(x) + \log D_3(x)] + \\ & \mathbb{E}_{z\sim p_z(z)}[\log(1 - D_1(G_1(z))) + \\ & \log(1 - D_2(G_1(z))) + \log(1 - D_3(G_2(z)))]\end{aligned} \quad (1)$$

Where:

- $\mathbb{E}_{x\sim p_{\text{data}}(x)}$ denotes the expectation over real data samples.
- $\mathbb{E}_{z\sim p_z(z)}$ refers to the expectation over the gradients for the spectral vectors sampled from the prior distribution $p_z(z)$.
- $D_1(x)$, $D_2(x)$, and $D_3(x)$ represent the output probabilities of the three discriminators for real data x.
- $D_1(G(z))$, $D_2(G_1(z))$, and $D_3(G_2(z))$ are the output probabilities of the three discriminators for synthetic data generated by the generators G.
- $V(D_1, D_2, D_3, G_1, G_2)$ is the value function of the minimax game, maximizing with respect to the discriminator parameters $(\theta_1, \theta_2, \theta_3)$ and minimizing with respect to the generators parameters (θ_h).

In the proposed GNN-GAN model, discriminator 3 (D_3) processes the generated fake data through a pre-trained GNN model, producing a graph representation structured as an N-tuple, with N denoting the number of multispectral bands and

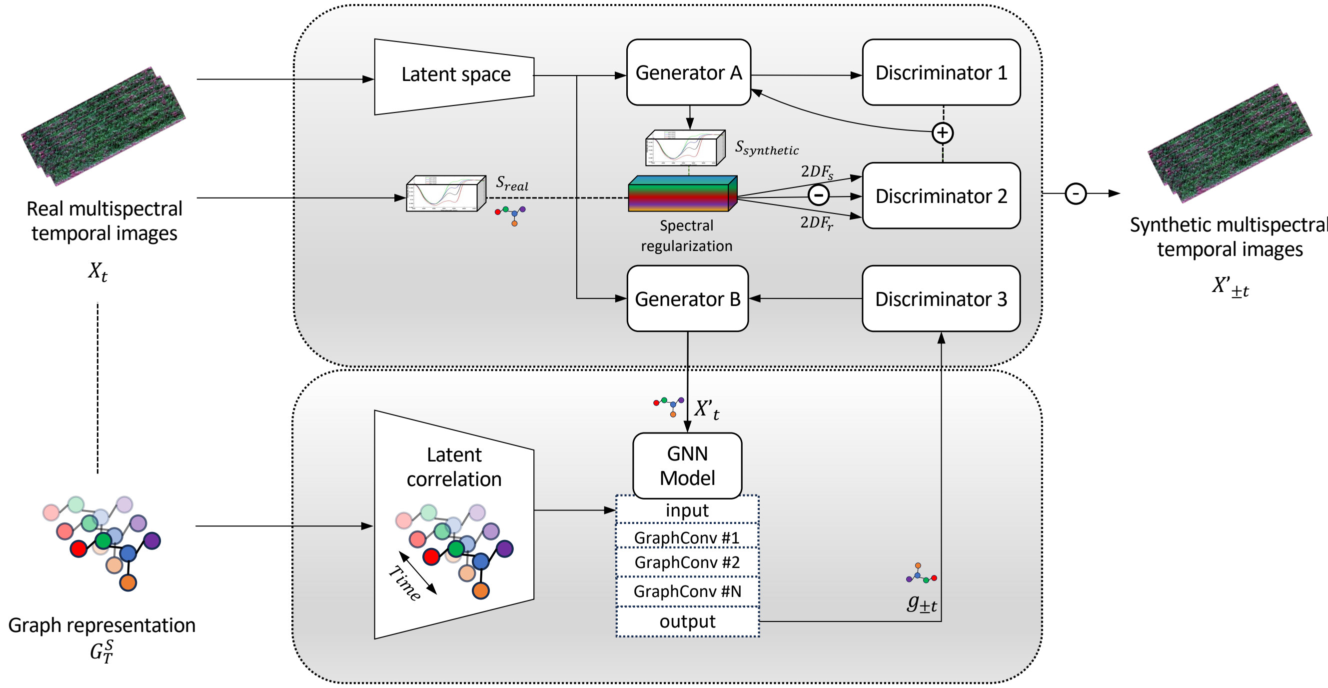

Fig. 2. An illustrated overview of the proposed GNN-GAN model showcasing the role of the third discriminator that evaluates the predicted multispectral temporal graph generated by a pre-trained GNN model. The GAN's generator components aim to minimize the disparity between the graph representing the real image at time $t \pm 1$ and the synthetic data generated at time $t \pm 1$, emphasizing the learning process required by the generator to achieve this minimization.

segmented into S segments representing nodes that group significant pixels. Notably, the resultant graph from the synthetic imagery is based on the output of the GNN model trained to return the next or previous probable graph according to a time t. Subsequently, the graph derived from the synthetic imagery is compared to the graph obtained from real data by calculating a distance L. This distance measurement, representing dissimilarity or similarity between graphs, utilizes the Euclidean distance computation based on node attributes or positions within the graph structure. The minimization problem for D_3 to minimize the Euclidean distance between corresponding node attributes or positions in the G_{real} and $G_{\text{synthetic}}$ graphs is represented as:

$$L = \underset{G_{\text{real}}, G_{\text{synthetic}}}{\text{minimize}} \sum_{i=1}^{N} \| G_{\text{real}}(i) - G_{\text{synthetic}}(i) \|^2 \quad (2)$$

Here, N represents the total number of nodes in the graph, and $G_{\text{real}}(i)$ and $G_{\text{synthetic}}(i)$ denote the attributes or positions of node i in the real and synthetic graphs, respectively. This minimization problem aims to reduce the dissimilarity between the structures represented by these graphs.

Graph-based representation and GNN Model. Utilizing a GNN for back and forecasting involves leveraging the graph representation of imagery to perform predictive tasks, capturing complex relationships within the image data.

Representing imagery as graphs, instead of spectral representations, for back and forecasting is justified by the inherent advantages of graph-based approaches in preserving spatial relationships, accommodating irregular data structures, and leveraging graph-based neural networks for predictive tasks. Graph representations maintain crucial spatial context by capturing pixel-to-pixel connections as nodes and edges, allowing for more accurate analysis of intricate relationships within the image – such pixel-to-pixel association is missed in spectral profiles. Thus, leveraging graph-based neural networks like GCNs facilitates forecasting by exploiting graph topology and capturing non-local relationships, enabling more nuanced and context-aware predictions. This choice empowered our predictive model to handle spatial dependencies and irregularities within the image data, potentially leading to more robust and accurate forecasting results.

To create the graph representation from multispectral imagery, the proposed methodology applies the SLIC (Simple Linear Iterative Clustering) algorithm [17] for superpixel segmentation. Superpixels are compact, contiguous regions in an image that group together pixels with similar properties, such as color or texture. The objective is to segment an input image into a specified number of superpixels, while additionally connecting neighboring superpixels in the graph by adding edges between them.

Let I denote the input dataset, where I has dimensions $(\text{length}, \text{height/width}, \text{channels/features})$. The SLIC (Simple Linear Iterative Clustering) algorithm aims to partition the dataset into K coherent segments or superpixels. It can be described as:

1. Initialization:

- Initialize K cluster centers or seeds $C_k = (l_k, a_k, b_k, ...)$, where k represents each cluster center and $l_k, a_k, b_k, ...$ denote feature values (e.g., in the case of images, l, a, b

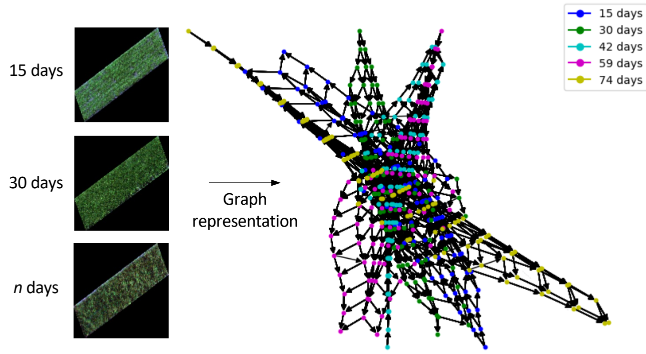

Fig. 3. Representation instance of our multispectral temporal dataset as a set of graphs.

might represent color channels).

2. Assignment Step:

- For each pixel p in the dataset I:
  - Compute the distance between p and each cluster center C_k using a spatial and feature space distance metric.
  - Assign p to the nearest cluster center based on these distances.

3. Update Step:

- Recalculate the cluster centers C_k by taking the mean of all pixels assigned to each cluster.
- Ensure adherence to compactness and boundary constraints.

The iterative process of assignment and update continues until convergence, minimizing the compactness and adherence to segment boundaries while considering the desired number of segments K. Figure 3 depicts the result of such transformation.

The formulaic representation of SLIC highlights the initialization of cluster centers, the assignment of pixels to clusters based on proximity, and the iterative refinement of cluster centers to achieve coherent segments within the dataset.

IV. Evaluation

The proposed method was evaluated from three different perspectives, described in the following sub-sections. First, we analyzed the performance of our method compared with other approaches and assessed the similarity between the synthetic temporal imagery and ground-truth data. Then, we explored if synthetic-based temporal vegetation indices would follow a real-world behavior. Last, we observed the impact of using our GNN-GAN integration in a scenario with limited temporal data and the performance of a downstream classification task when using the generated synthetic temporal data for training. The next sub-section describes the dataset utilized in our experiments.

A. Dataset

For training and evaluating the proposed GNN-GAN architecture, we utilized a multispectral imagery dataset of spring wheat collected on multiple dates at Chacabuco (-34.64; -60.46), Argentina. The data collection experiment consisted of growing three varieties of spring wheat, categorized as an intermediate-late-long cycle, intermediate cycle, and short cycle, in a total of 700 field plots. The plots were arranged in a grid pattern with 35 rows and 20 columns within the field. Each plot had dimensions of 1.2 x 5 meters and consisted of 7 rows of wheat plants. Spring wheat was planted on June 19th, 2021, and harvested on December 16th, 2021. The soil type at the site was classified as fine-silty, mixed, and thermic Typic Argiudoll according to USDA-Soil Taxonomy V. 2006. Throughout the growing season, uniform field management practices were implemented for all field plots. htInsecticide, herbicide, and fertilizer were applied as needed to ensure consistent plant health and growth. However, no fungicide was used in the experiment, as the focus was on studying the impact of the biotic stress caused by the occurrence of yellow rust disease.

UAV flights were conducted using the DJI Phantom 4 Pro as the primary aerial platform for collecting remotely sensed data from the standing crop. Equipped with a multispectral sensor capable of capturing five discrete spectral bands (blue, red, green, red edge, and near-infrared), the UAV was prepared for high-resolution imaging. A sunlight irradiance watcher was also connected to the sensor to ensure accurate measurements. The UAV was programmed to fly at a low altitude of 20 meters from the surface to achieve a high spatial resolution, resulting in a pixel resolution of 1.04 centimeters.

The aerial data collection was designed with a temporal dimension to monitor the crop at different stages of growth. The UAV conducted five flights throughout the growing stages, commencing with the first flight on August 30th, 2021, during the tillering stage, and concluding with the last flight on November 17th, 2021, during the flowering stage.

B. Similarity and performance analysis

To assess the similarity between the synthetic temporal imagery and ground-truth data, we trained the GNN model using samples from **three of the five** different time instants of our dataset, allowing $D_3(x'; G(x); \phi)$ to learn the latent correlations between these points. Then, we parameterized the time (*i.e.*, the number of days) for the training input of the GAN model, treating it as a feature received by $G(x)$. The exclusion of 2 instants (*i.e.*, 30 days and 42 days) from the training dataset was important to avoid a data leak to D_3. In this scenario, Figure 4 illustrates the R-squared values of mean spectral profiles derived from synthetic temporal imagery generated for an instant t from $t{+}1$, where t is limited to instances with comparable ground-truth data. The findings indicate that the model tends to predict higher values for bands 4 and 5 optimistically. Additionally, it exhibits suboptimal performance in capturing significant variations occurring over a short time span, such as the variations observed in spectral profiles between 40 and 42 days.

Expanding the time window to $t-3$ and computing the mean pixel values for all available ground-truth data, Figures 5 and 6 present the distribution of pixel values per band and the

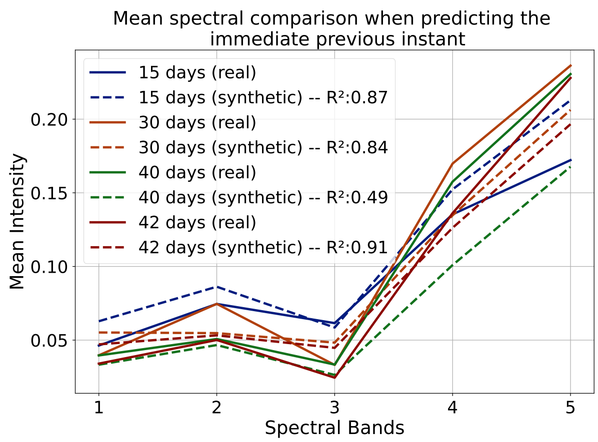

Fig. 4. The ground-truth dataset consists of multispectral imagery collected on five different dates. This chart depicts the prediction for the instant (t) $t-1$.

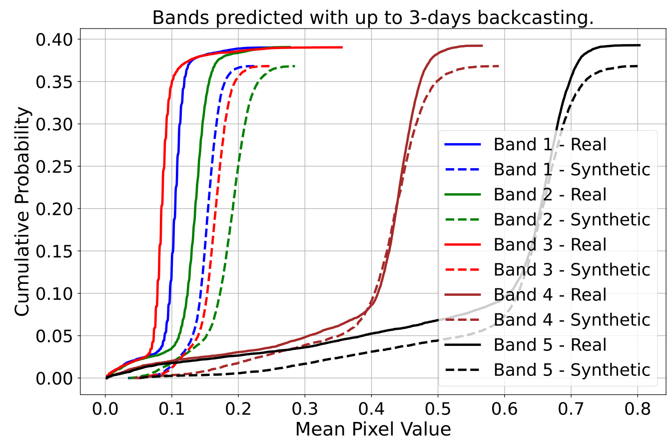

Fig. 6. Probability analysis of each band when backcasting the imagery for three days in the past using the proposed method. The model achieves a cumulative error of ∼20% when predicting the pixel values for bands 1–3.

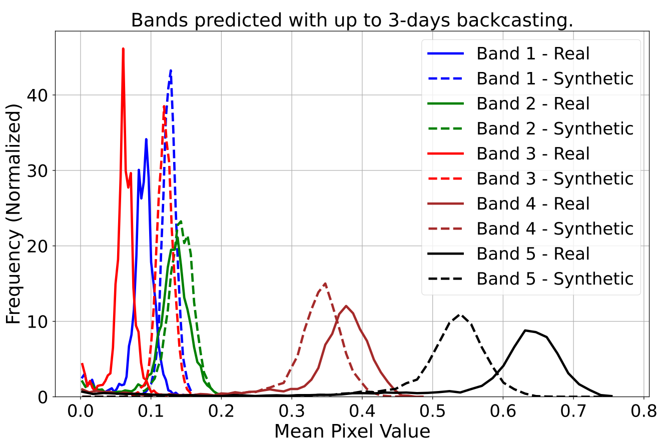

Fig. 5. Analysis of pixel value frequency when performing a backcasting for three days in the past using the proposed method.

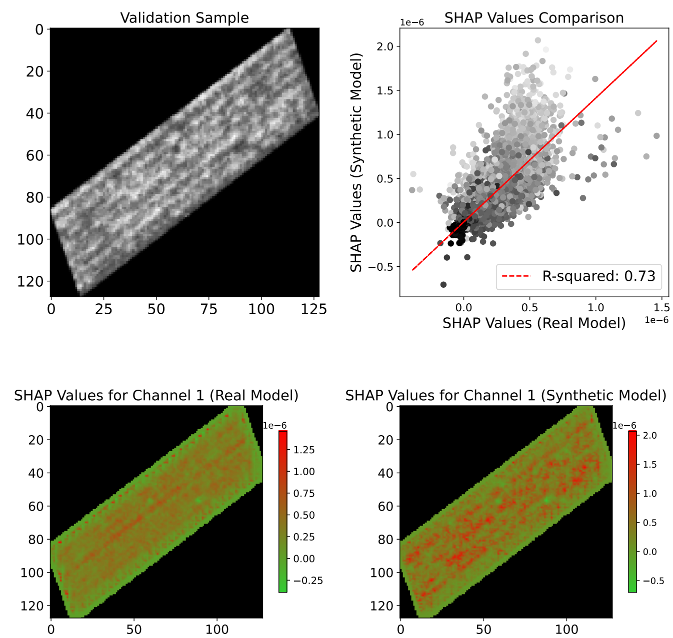

Fig. 7. At the top, the scatter plot compares SHAP values from the real and temporal synthetic-based CNN models for the same validation sample. Each point represents a pixel, with color indicating pixel intensity. The dashed red line represents the linear regression fit with an $R^2 = 0.73$, indicating a reasonable correlation between the SHAP values from both models. The two images at the bottom also indicate that the models identified the same pixel regions as diseased areas; however, the result for the model trained with temporal synthetic data (bottom-right) is spatially clearer.

probability of GMT accurately predicting each band's value. As anticipated, the model demonstrates superior performance in predicting pixel values for bands that exhibit minimal temporal variation.

In addition to verifying spectral profile similarity, we conducted a straightforward downstream pixel-based classification task, examining SHAP values from two CNN models: one exclusively trained on real samples and another trained on a combination of real and synthetic temporal data. Our objective is to detect diseased pixels within an unhealthy plot. Figure 7 illustrates the outcome of this experiment, wherein the CNN models accurately classify an unhealthy validation sample. The model trained with synthetic temporal samples offers a clearer spatial delineation of potential diseased plants, indicated by red pixels in the chart, compared to the more blurred region from the model trained solely on real imagery.

Finally, we evaluated the efficacy of our proposed model in comparison to similar models documented in the literature, namely TimeGAN, GT-GAN, and an LSTM-based GAN implementation. We conducted an assessment of synthetic samples generated by these models, analyzing three widely recognized metrics: Fréchet Inception Distance (FID), Bhattacharyya coefficient (BC), and Intersection over Union (IoU). All models were trained for 50 epochs, with 10 steps per epoch, using a learning rate of 0.0002 and the Adam optimizer. Additionally, while the metrics for PlantPlotGAN only consider predictions made for the same instant t, the other models attempt to predict plot appearances with a backcasting window of 7 days. The results of this evaluation are summarized in Table I. GTM demonstrates superior performance compared to other time series-based models in two key metrics: FID and

TABLE I
COMPARISON OF DIFFERENT MODELS BEING USED AS SPECTRAL-TEMPORAL PREDICTORS WHEN GENERATING TEMPORAL SYNTHETIC SAMPLES.

Models	FID ↓	BC ↓	IoU ↑
Baseline	0.21	0.30	0.04
GTM (*ours*)	**0.19**	**0.54**	0.20
TimeGAN [13]	0.22	0.63	0.19
GT-GAN [14]	0.31	0.68	**0.24**
LSTM	0.91	0.73	0.03
PlantPlotGAN [18]	0.07	0.51	0.12

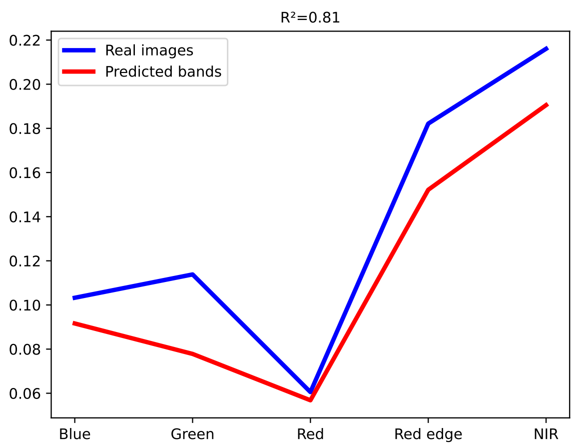

Fig. 8. Comparison of mean spectral profile predicted using an LSTM-based GAN reference model for an instant $t-3$.

BC. However, it does not achieve the highest Intersection result from the analysis. We attribute GTM's superior performance in FID and BC to the high-resolution temporal data dependency present in the other models, as well as the physics-constrained spectral value generation implemented in GTM (as discussed in Section III). On the other hand, the pixel cumulative probability for bands 4 and 5 obtained from the samples generated by GTM negatively impacted its IoU result. It's worth noting that the baseline for these experiments consists of rotated samples, all considering the same instant t.

C. Feature analysis between temporal synthetic and real imagery

We evaluated whether the temporal imagery generated from GMT could exhibit behavior similar to real imagery when analyzed through the perspective of vegetation indices. We performed feature engineering to extract 43 vegetation indices (VIs) selected from the literature [19] (see Table II). From these 43 VIs, we arbitrarily selected five indices and analyzed them using a window size of up to 7 days during backcasting. We then compared the results with our ground truth data as follows:

- **NDVI** (Normalized Difference Vegetation Index): NDVI measures the amount of vegetation present in an area. It's calculated from the difference between near-infrared (NIR) and red light reflectance. Higher NDVI values generally indicate healthier and more abundant vegetation.

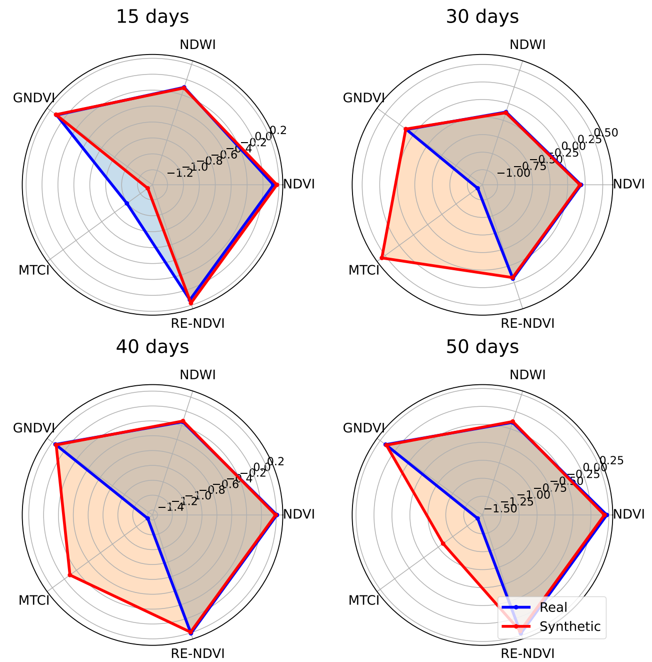

Fig. 9. Comparison of vegetation indices for each plant growth stage derived from real and synthetic data with a backcasting window of up to 7 days.

- **NDWI** (Normalized Difference Water Index): NDWI is used to detect the presence of water in vegetation. It's calculated from the difference between NIR and green light reflectance. Higher NDWI values typically indicate the presence of water.
- **GNDVI** (Green Normalized Difference Vegetation Index): GNDVI is similar to NDVI but uses green light instead of red light. It's also a measure of vegetation density and health.
- **MTCI** (Meris Terrestrial Chlorophyll Index): MTCI is a vegetation index sensitive to chlorophyll content in leaves and helps in assessing plant health and stress.
- **RE-NDVI** (Red Edge Normalized Difference Vegetation Index): RE-NDVI is a modified version of NDVI that uses the red-edge bands of satellite imagery. It's particularly useful for monitoring vegetation stress and changes in plant phenology.

These indices offer a comprehensive view of the physiological state and health status of the plants.

Figure 9 shows the result of this analysis. The temporal synthetic plots were able to achieve high similarity with the real ones, which suggests that – even for dates without ground-truth data – our model is capable of generating samples that reproduce the appearance (from the spectral and spatial perspective) of real imagery.

D. Impact of synthetic temporal data on prediction models.

We also designed an experiment to analyze whether synthetic temporal data could improve the accuracy of prediction models when analyzing VIs (cf. Table II). In this experiment, we generated synthetic temporal imagery for each date of the ground-truth data (mostly for the minority class – healthy

TABLE II
VEGETATION INDEX FORMULAS AND REFERENCES - PART 1

No.	Index Names	Index Formula
1	Normalized difference vegetation index (NDVI)	$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}}$ [20]
2	Green normalized difference vegetation index (GNDVI)	$\mathrm{GNDVI} = \frac{\mathrm{NIR}-\mathrm{G}}{\mathrm{NIR}+\mathrm{G}}$ [21]
3	Ratio vegetation index (RVI)	$\mathrm{RVI} = \frac{\mathrm{NIR}}{\mathrm{R}}$ [22]
4	Green chlorophyll index (GCI)	$\mathrm{GCI} = \frac{\mathrm{NIR}}{\mathrm{G}} - 1.0$ [23]
5	Red-green ratio vegetation index (RGVI)	$\mathrm{RGVI} = \frac{\mathrm{R}}{\mathrm{G}}$ [24]
6	Difference vegetation index (DVI)	$\mathrm{DVI} = \mathrm{NIR} - \mathrm{R}$ [25]
7	Soil-adjusted vegetation index (SAVI)	$\mathrm{SAVI} = \left(\frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}+0.5}\right) \times (1.0 + 0.5)$ [26]
8	Modified soil-adjusted vegetation index (MSAVI)	$\mathrm{MSAVI} = 0.5 \times \left(2.0 \times \mathrm{NIR} + 1.0 - \sqrt{(2.0 \times \mathrm{NIR} + 1.0)^2 - 8.0 \times (\mathrm{NIR} - \mathrm{R})}\right)$ [27]
9	Optimized soil-adjusted vegetation index (OSAVI)	$\mathrm{OSAVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}+0.16}$ [28]
10	Renormalized difference vegetation index (RDVI)	$\mathrm{RDVI} = \sqrt{\frac{(\mathrm{NIR}-\mathrm{R})^2}{\mathrm{NIR}+\mathrm{R}}}$ [29]
11	Triangular vegetation index (TVI)	$\mathrm{TVI} = 60.0 \times (\mathrm{NIR} - \mathrm{G}) - 100.0 \times (\mathrm{R} - \mathrm{G})$ [30]
12	Transformed soil-adjusted vegetation index (TSAVI)	$\mathrm{TSAVI} = \frac{a \times (\mathrm{NIR} - a \times \mathrm{R} - b)}{a \times \mathrm{NIR} + \mathrm{R} - a \times b}$ [31]
13	Perpendicular vegetation index (PVI)	$\mathrm{PVI} = \frac{\mathrm{NIR} - a \times \mathrm{R} - b^{0.5}}{1+a^2}$ [32]
14	Soil-adjusted vegetation index 2 (SAVI_2)	$\mathrm{SAVI}_2 = \frac{\mathrm{NIR}}{\mathrm{R} - \frac{b}{a}}$ [33]
15	Adjusted transformed soil-adjusted vegetation index (ATSAVI)	$\mathrm{ATSAVI} = \frac{a \times (-a \times \mathrm{R} - b)}{a \times \mathrm{NIR} + \mathrm{R} - a \times b + X \times (1 + a^2)}$ [34]
16	Normalized difference water index (NDWI)	$\mathrm{NDWI} = \frac{\mathrm{G}-\mathrm{NIR}}{\mathrm{G}+\mathrm{NIR}}$ [35]
17	Normalized total pigment to chlorophyll a ratio index (NPCI)	$\mathrm{NPCI} = \frac{\mathrm{R}-\mathrm{B}}{\mathrm{R}+\mathrm{B}}$ [36]
18	Simple ratio pigment index (SRPI)	$\mathrm{SRPI} = \frac{\mathrm{B}}{\mathrm{R}}$ [36]
19	Ratio vegetation index (RVI_2)	$\mathrm{RVI}_2 = \frac{\mathrm{NIR}}{\mathrm{G}}$ [37]
20	Modified chlorophyll absorption ratio index (MCARI)	$\mathrm{MCARI} = (\mathrm{RE} - \mathrm{R} - 0.2 \times (\mathrm{RE} - \mathrm{G})) \times (\mathrm{RE}/\mathrm{R})$ [38]
21	Modified chlorophyll absorption ratio index 1 (MCARI_1)	$\mathrm{MCARI}_1 = 1.2 \times (2.5 \times (\mathrm{NIR} - \mathrm{R}) - 1.3 \times (\mathrm{NIR} - \mathrm{G}))$ [39]
22	Modified chlorophyll absorption ratio index 2 (MCARI_2)	$\mathrm{MCARI}_2 = 1.5 \times (2.5 \times (\mathrm{NIR} - \mathrm{R}) - 1.3 \times (\mathrm{NIR} - \mathrm{G})) \times ((2.0 \times \mathrm{NIR} + 1)^2 - (6.0 \times \mathrm{NIR} - 5.0 \times \mathrm{R}) - 0.5)$ [39]
23	Modified triangular vegetation index 1 (MTVI_1)	$\mathrm{MTVI}_1 = 1.2 \times (1.2 \times (\mathrm{NIR} - \mathrm{G}) - 2.5 \times (\mathrm{R} - \mathrm{G}))$ [39]
24	Modified triangular vegetation index 2 (MTVI_2)	$\mathrm{MTVI}_2 = 1.5 \times (1.2 \times (\mathrm{NIR} - \mathrm{G}) - 2.5 \times (\mathrm{R} - \mathrm{G})) \times ((2 \times \mathrm{NIR} + 1)^2 - (6.0 \times \mathrm{NIR} - 5.0 \times \mathrm{R}) - 0.5)$ [39]
25	Ratio of Modified chlorophyll absorption ratio index and Modified triangular vegetation index 2 ($\mathrm{RMCARI}_{\mathrm{MTVI}_2}$)	$\mathrm{RMCARI}_{\mathrm{MTVI}_2} = \frac{(\mathrm{RE}-\mathrm{R}-0.2\times(\mathrm{RE}-\mathrm{G}))\times(\mathrm{RE}/\mathrm{R})}{1.5\times(1.2\times(\mathrm{NIR}-\mathrm{G})-2.5\times(\mathrm{R}-\mathrm{G}))\times((2\times\mathrm{NIR}+1)^2-(6.0\times\mathrm{NIR}-5.0\times\mathrm{R})-0.5)}$ [40]
26	Enhanced vegetation index (EVI)	$\mathrm{EVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+6.0\times\mathrm{R}-7.5\times\mathrm{B}+1.0}$ [41]
27	Datt's chlorophyll content index (DATT)	$\mathrm{DATT} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}-\mathrm{R}}$ [42]
28	Normalized difference cloud index (NDCI)	$\mathrm{NDCI} = \frac{\mathrm{RE}-\mathrm{G}}{\mathrm{RE}+\mathrm{G}}$ [43]
29	Plant senescence reflectance index (PSRI)	$\mathrm{PSRI} = \frac{\mathrm{R}-\mathrm{G}}{\mathrm{RE}}$ [44]
30	Structure insensitive pigment index (SIPI)	$\mathrm{SIPI} = \frac{\mathrm{NIR}-\mathrm{B}}{\mathrm{NIR}+\mathrm{R}}$ [45]
31	Spectral polygon vegetation index (SPVI)	$\mathrm{SPVI} = 0.4 \times 3.7 \times (\mathrm{NIR} - \mathrm{R}) - 1.2 \times \lvert \mathrm{G} - \mathrm{R} \rvert$ [46]
32	Transformed chlorophyll absorption in reflectance index (TCARI)	$\mathrm{TCARI} = 3.0 \times ((\mathrm{RE} - \mathrm{R}) - 0.2 \times (\mathrm{RE} - \mathrm{G}) \times (\mathrm{RE}/\mathrm{R}))$ [47]
33	Ratio of TCARI and OSAVI ($\mathrm{R}_{\mathrm{TCARI}_{\mathrm{OSAVI}}}$)	$\mathrm{R}_{\mathrm{TCARI}_{\mathrm{OSAVI}}} = \frac{3.0\times((\mathrm{RE}-\mathrm{R})-0.2\times(\mathrm{RE}-\mathrm{G})\times(\mathrm{RE}/\mathrm{R}))}{(\mathrm{NIR}-\mathrm{R})/(\mathrm{NIR}+\mathrm{R}+0.16)}$ [47]
34	Red edge relative index (RERI)	$\mathrm{RERI} = \frac{\mathrm{RE}-\mathrm{R}}{\mathrm{NIR}}$ [48]
35	Normalized difference red edge index (NDRE)	$\mathrm{NDRE} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}+\mathrm{RE}}$ [49]
36	MERIS terrestrial chlorophyll index (MTCI)	$\mathrm{MTCI} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{RE}-\mathrm{R}}$ [50]
37	Enhanced vegetation index 2 (EVI_2)	$\mathrm{EVI}_2 = 2.5 \times \left(\frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+2.4\times\mathrm{R}+1.0}\right)$ [51]
38	Red edge chlorophyll index (RECI)	$\mathrm{RECI} = \frac{\mathrm{NIR}}{\mathrm{RE}} - 1$ [23]
39	Normalized excess green index (NEXG)	$\mathrm{NEXG} = \frac{2\times\mathrm{G}-\mathrm{R}-\mathrm{B}}{\mathrm{G}+\mathrm{R}+\mathrm{B}}$ [52]
40	Normalized green-red difference index (NGRDI)	$\mathrm{NGRDI} = \frac{\mathrm{G}-\mathrm{R}}{\mathrm{G}+\mathrm{R}}$ [25]
41	Enhanced normalized difference vegetation index (ENDVI)	$\mathrm{ENDVI} = \frac{\mathrm{NIR}+\mathrm{G}-2.0\times\mathrm{B}}{\mathrm{NIR}+\mathrm{G}+2.0\times\mathrm{B}}$ [53]
42	Anthocyanin reflectance index 2 (ARI_2)	$\mathrm{ARI}_2 = \mathrm{NIR} \times \left(\frac{1.0}{\mathrm{G}} - \frac{1.0}{\mathrm{RE}}\right)$ [54]
43	Carotenoid reflectance index 2 (CRI_2)	$\mathrm{CRI}_2 = \frac{1.0}{\mathrm{G}} - \frac{1.0}{\mathrm{RE}}$ [55]

plots), always considering the immediate previous instant, and extracted the 43 VIs. Afterward, we verified if the classification of plant disease (annotated at the last date from the ground truth) would improve using early-stage data (real and synthetic). We also compared the results with those obtained in a previous work [18], which focused solely on balancing the dataset (i.e., healthy and unhealthy) using synthetic data. The validation set was the same in all scenarios.

The results of this experiment indicate that our GNN-GAN model is capable of generating realistic temporal synthetic data and improving the accuracy (cf. Equation 3) of a regular Convolutional Neural Network (CNN) by up to 53% (cf. Figure 10) at the earliest stages of plant growth when predicting plant disease. Such finding suggests a significant benefit of performing scene reconstruction or temporal data augmentation using GANs and GNN-based generative techniques.

$$\text{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \times 100\% \tag{3}$$

V. DISCUSSION

The evaluation of the proposed method encompasses various aspects, each shedding light on different facets of its performance and potential applications. Here, we discuss the findings and implications of each evaluation perspective.

The comparison of our method with existing approaches forms the cornerstone of our evaluation. By assessing the

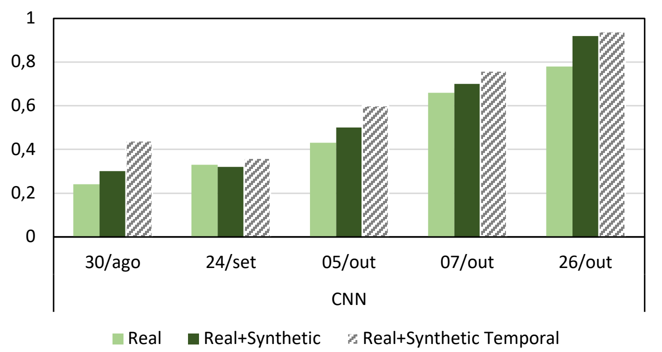

Fig. 10. Analyzing the accuracy of a CNN-based classification forecast model using different datasets for training the predictor, including the training set that combines the synthetic temporal data generated by the proposed method. The validation set used was the same in all scenarios. As the chart depicts, the utilization of synthetic temporal data helped improve accuracy in the early stages of plant growth.

performance against other methods and examining the similarity between synthetic temporal imagery and ground-truth data, we gauged the efficacy and realism of our approach. Notably, excluding certain time instants during training to prevent data leakage underscores our method's robustness in capturing temporal dynamics without relying on specific data points.

The statistical analysis, including FID, Battacharayya, and intersection metrics (as presented in Table I), offers a quantitative assessment of the performance relative to other methods. These metrics provide valuable insights into the fidelity and resemblance of the generated imagery compared to ground truth, enabling a comprehensive evaluation of our approach's efficacy.

Feature analysis, particularly concerning vegetation indices, is crucial for understanding the behavior of synthetic imagery relative to real-world data. The comprehensive exploration of 43 VIs, coupled with the comparison of selected indices using a range of window sizes during backcasting, could demonstrate the capability of our method to replicate real-world vegetation dynamics.

The comparison of indices such as NDVI, NDWI, GNDVI, MTCI, and RE-NDVI, along with the radar chart visualization presented in Figure 9, illustrates the high degree of similarity between synthetic and real imagery. This suggests that our model effectively captures vegetation dynamics over time, which is essential for various applications such as agriculture monitoring and environmental assessment. The low accuracy in predicting MTCI is explainable by the bands used to calculate it. MTCI's bands have the highest distance when analyzing the mean pixel value (cf. Figure 5). Future work could focus on verifying the optimum model size to train the GAN components in order to minimize such distances.

The experiment assessing the impact of synthetic temporal data on prediction models offers compelling evidence of the utility and effectiveness of our approach. The significant improvement in classification accuracy, especially in the early stages of plant growth, underscores the potential of synthetic temporal data in enhancing the performance of prediction models. This finding has far-reaching implications for applications such as disease detection and yield forecasting, where accurate and timely data play a crucial role.

Another open issue is in splitting the multispectral data into different generator streams in the GAN architecture, with the hypothesis that each generator could perform better if associated with a specific multispectral band. Such a hypothesis is out of the scope of the current work.

A. Limitations

We tackled the challenges of scene reconstruction and temporal data augmentation by leveraging the latent space employed by Generative Adversarial Networks (GANs), with a primary focus on improving the accuracy of subsequent classification tasks using synthetic temporal data. To reduce computational demands, our input layer comprises randomly generated vectors of size $N100$. Future research could explore the feasibility of inputting specific multispectral plot data and reconstructing its historical context to generate synthetic imagery for a desired past time frame, bearing in mind the high computational resources required for such an approach.

Another aspect requiring attention is the evaluation of additional real-world temporal datasets. This would help demonstrate the feasibility and generalization of the proposed method across diverse scenarios.

VI. Conclusion

The study systematically introduced a technique based on Graph Neural Networks (GNN) and Generative Adversarial Networks (GAN) to generate temporal data for past contexts. The investigation focused on the benefits of utilizing synthetic temporal data for the early detection of infectious diseases, specifically yellow rust in wheat, using aerial remote sensing imagery as the primary data source.

From the findings and discussions presented in Section V, the following conclusions were drawn:

1) The approach of representing plots as graphs, as opposed to relying solely on spectral profiles, shows promise in preserving both spatial and spectral information from the imagery.
2) Critical features extracted from imagery had the same importance in models trained with real data and with a combination of real and synthetic temporal data.
3) The integration of synthetic temporal data has the potential to enhance early disease detection by up to 50% compared to models trained exclusively on real data or balanced synthetic imagery.
4) The validation study revealed that the classification models trained with real imagery alone or a combination of real and synthetic temporal imagery utilized the same regions (i.e., pixels) of the control group.

Interested readers are strongly encouraged to acknowledge the study's limitations as detailed in Section V-A, which include the necessity for a plot-specific temporal characterization and a comprehensive investigation into the tradeoff between spectral-based and graph-based approaches.

The achieved results open avenues to future work that might explore the opportunities of performing: i) scene reconstruction utilizing the proposed method in other scenarios (e.g., animal trajectory, urban development); ii) data balancing; and iii) outlier detection in scarce temporal multispectral data contexts.

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